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AC LAMP BLINKER
Using Timer NE555

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There are at least two popular applications of blinking a string of 110V/230V AC incandescent lamps. The first is audio-visual warning at construction sites or where repairing is going on. The second is to express joy and excitement during holidays like New Year's Day.

Here we use a very simple and low-cost timer NE555 to switch on and off two output loads alternately for audio and visual indications. You can achieve this by using a bipolar-transistor-based NE555 or CMOS-based LMC555.

This circuit can be made to blink AC lamps at a low frequency, or switch on and off electrical loads connected to the mains at a low speed. In order to reduce the RF emissions, switching is done only at zero crossings of the mains AC voltage.

The switching frequency is selected according to the resistor and capacitor components connected to NE555. It can be varied using 100-kilo-ohm potmeter VR1.

PARTS LIST

Semiconductors:	
IC1	- 7809, 9V regulator
IC2	- NE555 timer
IC3, IC4	- MOC3043 optoisolator
D1, D2	- 1N4007 rectifier diode
BR1	- 1A bridge rectifier
Triac1, Triac2	- BTA06 triac
LED1-LED6	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon, unless stated otherwise):	
R1	- 100-ohm, 1-watt
R2	- 1-kilo-ohm
R3, R5	- 10-kilo-ohm
R4	- 33-kilo-ohm
R6-R8	- 330-ohm
R9, R10, R12, R15	- 220-ohm
R11, R13, R16	- 100-kilo-ohm
R14, R17	- 100-ohm, 0.5-watt
VR1	- 100-kilo-ohm potmeter
Capacitors:	
C1, C8-C11	- 0.1µF, 400V polyester
C2, C5	- 0.1µF ceramic disk
C3	- 470µF, 25V electrolytic
C4, C6	- 100µF, 16V electrolytic
C7	- 0.01µF ceramic disk
Miscellaneous:	
CON1-CON3	- 2-pin terminal connector
F1	- 0.1A fuse with holder
F2, F3	- 1A fuse with holder
PZ1, PZ2	- Piezo-buzzer
S1-S4	- On/off switch
Load1, Load2	- 230V AC, 60W bulb/lamp
X1	- 230V AC primary to 12V, 250mA secondary transformer
	- 230V AC mains power supply
	- Heat-sinks for triacs
	- Mains power cord

Circuit and working

Circuit diagram of the AC lamp blinker is shown in Fig. 1. It is built around a step-down transformer (X1), 9V voltage regulator 7809 (IC1), timer NE555 (IC2), two MOC3043 optoisolators (IC3 and IC4), two BTA06 triacs (Triac1 and Triac2), two 1N4007 diodes (D1 and D2), six LEDs (LED1 through LED6), two piezo-buzzers (PZ1, PZ2), two 230V AC, 60W bulbs (Load1 and Load2), and a few other components.

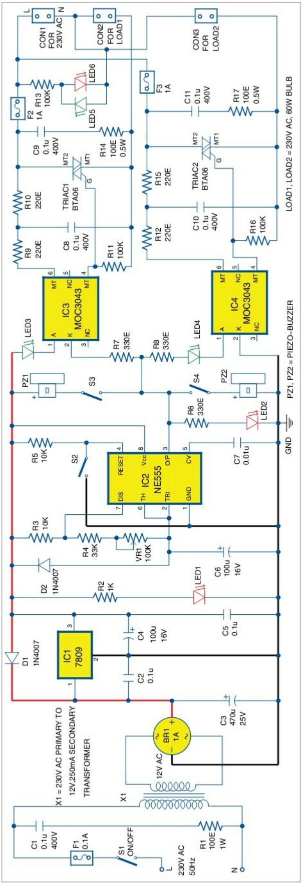

Fig. 1: Circuit diagram of AC lamp blinker using timer NE555